$$\underset{\text{sodium dichromate (orange red)}}{Na_2Cr_2O_7} + 2KCl \longrightarrow \underset{\text{potassium dichromate (orange red)}}{K_2Cr_2O_7} + 2NaCl$$

Physical properties:

Potassium dichromate is an orange red crystalline solid which melts at 671K and it is moderately soluble in cold water, but very much soluble in hot water. On heating it decomposes and forms Cr_2O_3 and molecular oxygen. As it emits toxic chromium fumes upon heating, it is mainly replaced by sodium dichromate.

$$4\ \underset{\text{potassium dichromate}}{K_2Cr_2O_7} \xrightarrow{\Delta} 4\ \underset{\text{potassium chromate}}{K_2CrO_4} + 2\ \underset{\text{chromium(III) oxide}}{Cr_2O_3} + 3O_2 \uparrow$$

Structure of dichromate ion:

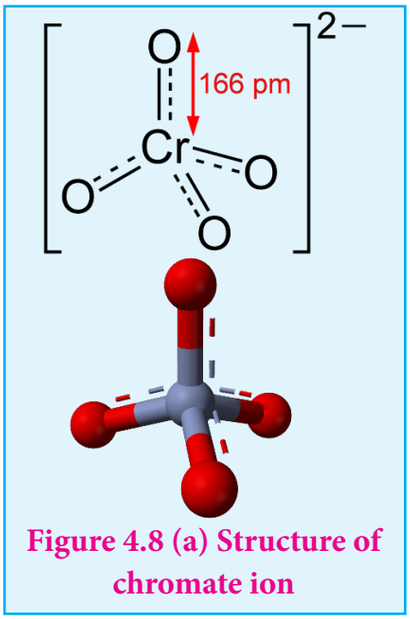


Figure 4.8 (a) Structure of chromate ion

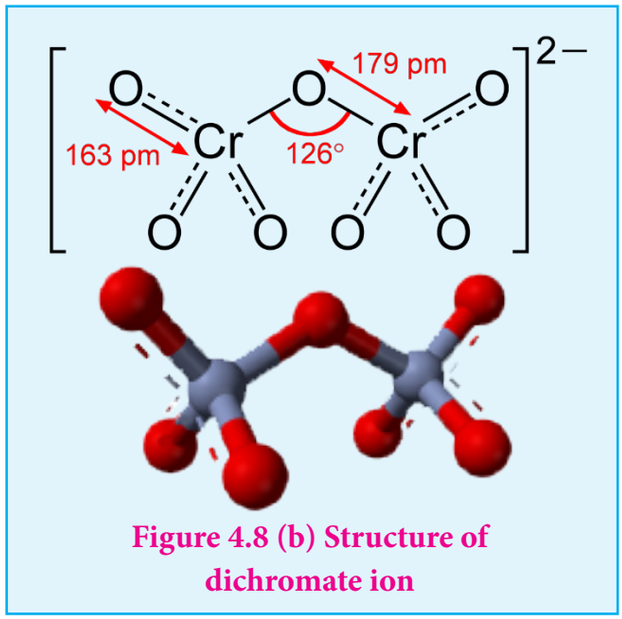


Figure 4.8 (b) Structure of dichromate ion

Both chromate and dichromate ion are oxo anions of chromium and they are moderately strong oxidizing agents. In these ions chromium is in +6 oxidation state. In an aqueous solution, chromate and dichromate ions can be interconvertible, and in an alkaline solution chromate ion is predominant, whereas dichromate ion becomes predominant in acidic solutions. Structures of these ions are shown in the figure.

Chemical properties:

1. Oxidation

Potassium dichromate is a powerful oxidising agent in acidic medium. Its oxidising action in the presence of H^+ ions is shown below. You can note that the change in the oxidation state of chromium from Cr^{6+} to Cr^{3+}.Its oxidising action is shown below.

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7\ H_2O$$

The oxidising nature of potassium dichromate (dichromate ion) is illustrated in the following examples.

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